

Operational Interoperability Module (OIM)

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Canonical Definition

Operational Interoperability Module (OIM) defines the structural conditions under which autonomous operational systems exchange information, coordinate activities, and perform interoperable operations while preserving governance integrity, operational consistency, and trustworthy collaboration.

OIM governs operational interoperability.

The module establishes the governance conditions required to ensure that autonomous operational actors remain capable of operating together despite differences in implementation, architecture, or operational environment.

Compatibility enables cooperation.

Interoperability enables operation.

OIM governs that interoperability.

Module Operational Role

OIM defines the operational interoperability layer of ACPS.

Its role is to establish and preserve interoperable autonomous

Module Operational Space

OIM governs:

- operational interoperability,
 - interoperable autonomous operations,
 - cross-system interaction,
 - operational integration,
 - governed interoperability,
 - collaborative operational exchange,
 - interoperability governance,
 - autonomous ecosystem integration.
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Module Function

The module applies wherever autonomous operational actors must exchange information or cooperate across different systems, platforms, or operational environments.

Minimum Implementation Framework

1. Define the Interoperability Object

Identify which autonomous operational systems require interoperable operation.

2. Define Interoperability Conditions

Establish governance conditions governing interoperable interaction and operational cooperation.

3. Define Interoperability Failure Logic

Identify conditions where interoperability becomes incompatible, interrupted, inconsistent, or governance-invalid.

4. Define Governance Response

Define governance actions for interoperability validation, operational adaptation, coordination adjustment, escalation, or recovery.

5. Preserve Interoperability Traceability

Preserve interoperability records, governance decisions, operational evidence, interaction history, and supporting documentation.

Use Case 1 — Autonomous Smart Port

Scenario

Autonomous cranes, transport vehicles, and logistics systems coordinate cargo handling across a large maritime terminal.

Application

OIM governs interoperable communication and operational cooperation between independently developed autonomous systems.

Result Port operations remain coordinated, efficient, and continuously governed.

Use Case 2 — Multi-Vendor Autonomous Manufacturing

Scenario

An industrial production line combines autonomous robots supplied by different manufacturers.

Application

OIM governs interoperable operation across heterogeneous autonomous platforms.

Result

The manufacturing environment functions as a unified, interoperable, and governance-compliant autonomous ecosystem.

Canonical Closing Statement

Governable compatibility requires governable interoperability.
Operational Interoperability ensures that autonomous operational systems collaborate consistently, reliably, and under continuous governance across diverse operational environments.

Related Documents

→ [Autonomous Compatibility Standard \(ACPS\)](https://originopenfoundation.org/content/a/asga-acps-oim-1.html).

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